Kappa Foundation — Initial Decision Record

BFIN WS2026 Investments · Group Assignment · Decision date 31 Dec 2022 · Submitted 22 Oct 2026

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1. Decision

- **Fixed payments** USD $2.4 million on December 31st 2024 and USD $3.6 million on December 31st 2027. The funds for these fixed payments were matched exactly to the cash flow from the sale of 2,157 Beta + 3,479 Delta bonds at a cost of USD $5,420,974. The value today of those future payments is USD $5,202,169. In addition to the bond portion, the coupon income from the Delta Bonds issued in 2023 & 2024 will be used to fund the USD $2.4 million in payments due on December 31st 2024. The remaining 2025 – 2026 coupons will be placed into the growth component.
- **Remainder** Remaining USD $6,479,026 (the remainder after creating a USD $100k cash reserve): This amount will be allocated as follows:
    + 49% to a growth portfolio of ten U.S. Stocks + Two U.S. Equity Exchange Traded Funds ("ETF's").
    + The balance (51%) will be invested in approximately 3,175 Epsilon Bonds, which mature on December 31st 2030 (which is our goal date). If we had only invested in "safe" bonds, the estimated amount available on December 31st 2022 would have been less than the target amount of USD $8M.
- **Growth portfolio:** Max Sharpe optimization with CAPM expected returns (beta x ERP), using a single index covariance matrix and weight limits. Estimated Return = 8.83%. Estimated Volatility = 15.16%. Estimated Beta = .91. Approximate probability of reaching an estimated USD $8M real by 2030 ≈ 67%. Stress Loss = 14.8%. One-Year 95% Value-at-Risk = 3.9% net-of-payments.

2. Candidate universe and eligibility (evidence: Yahoo trade on 30 Dec 2022 with volume > 0; latest SEC filing ≤ 31 Dec 2022)

Ticker	Name	Sector	Close 30 Dec 22	SEC filing	Ticker	Name	Sector	Close 30 Dec 22	SEC filing
MSA	MSA Safety	Industrials	144.19	10-Q 2022-10-27	NWN	Northwest Natural Hldg	Utilities	47.59	10-Q 2022-11-08
KAI	Kadant	Industrials	177.63	10-Q 2022-11-09	CHE	Chemed	Health Care	510.43	10-Q 2022-11-02
GEF	Greif (Cl. A)	Materials	67.06	10-K 2022-12-16	NSIT	Insight Enterprises	Info. Tech.	100.27	10-Q 2022-11-03
IOSP	Innospec	Materials	102.86	10-Q 2022-11-09	XMLV	Invesco S&P MidCap Low Vol ETF	diversified	53.49	NPORT-P 2022-12-30
FLO	Flowers Foods	Cons. Staples	28.74	10-Q 2022-11-10	RWL	Invesco S&P 500 Revenue ETF	diversified	73.94	NPORT-P 2022-12-30
CALM	Cal-Maine Foods	Cons. Staples	54.45	10-Q 2022-12-28	SPY	SPDR S&P 500 ETF (benchmark)	diversified	382.43	NPORT-P 2022-11-28
IDA	IDACORP	Utilities	107.85	10-Q 2022-11-03					

Our selection criteria included the following:

- Listed on the United States stock exchanges;
- Active trading history during the years 2015 – 2022; and,
- Mostly mid-capitalization companies which paid dividends.

In selecting candidates for inclusion within each of six sectors, we selected two candidate stocks per sector wherever possible (e.g., Financial Services, Healthcare etc.) to provide broad representation within each sector. We initially identified CSG Systems (CSGS) as one of the initial candidates however due to lack of data from Yahoo Finance we could not select it (it appears to be delisted); therefore we substituted NSIT. These two biases mentioned above represent reasons why we did not utilize historical mean returns. Our benchmark was established by utilizing SPY (S&P 500 Index Fund) and measuring their performance via simple buy-and-hold total return methodology, which provided a reasonable proxy for investment returns based upon historical data.

3. Estimation window and method

This section provides additional details about the estimation time frame and methodology. The monthly Total Returns (Adj Close) for the period January 2016 through December 2022 (84 months) were used. Excess returns for this period were compared against those of the French Return Free. In addition to the entire calendar year, the dataset included both the 2020 Crash and the 2022 Rate Shock. Two sub-periods also existed, including 2016-2019 and 2020-2022. The two periods have differing weights based upon the percentage point variation in each sub-period. These variations range from less than five percent (≤5pp) which is equivalent to a 5-7 percent turnover (sub-window turnovers). Bonds were modeled using their Yield To Maturity (YTM), bootstrapped yield curves (linear spot check; flat forward rates); duration/convexity and Treasury-size rate scenarios. Based on these parameters, dedication was able to perform better than Gamma + epsilon immunization during a yield curve twist. Rejecting previous methods included: Eta (sample-covariance-based historical-max-Sharpe, no constraints: 6 asset class names; 85 percent in CHE/MSA/KAI) and the sample-covariance version that did not include any bounds (therefore did not do as well out-of-sample).

4. Initial dollar allocation, 31 Dec 2022 (whole bonds/shares at quoted / closing prices, incl. trading costs)

Holding	Units	USD	% fund	Holding	Units	USD	% fund
Beta (bond)	2,157	1,990,436	16.59%	CALM	582	31,722	0.26%
Delta (bond)	3,479	3,430,538	28.59%	IDA	1,285	138,726	1.16%
Epsilon (bond)	3,175	3,304,254	27.54%	NWN	1,928	91,845	0.77%
MSA	1,359	196,150	1.63%	CHE	199	101,677	0.85%
KAI	767	136,378	1.14%	NSIT	1,029	103,281	0.86%
GEF	1,574	105,658	0.88%	XMLV	18,287	978,465	8.15%
IOSP	1,042	107,287	0.89%	RWL	15,023	1,111,134	9.26%
FLO	2,487	71,548	0.60%	Cash		100,900	0.84%
Total		12,000,000	100.00%	of which growth portfolio		3,173,872	26.45%

5. Material assumptions (sources and reasoning: theory_and_sources/B)

Assumption	Value	Basis
Inflation 2023–2030	2.3% (stress 3.5%)	Treasury breakeven 30 Dec 2022: 5y 2.33%, 10y 2.30%; Fed goal 2%
Equity risk premium	5.0%	Damodaran: historical 1928–2022 5.1% (geom.), implied Jan 2023 5.94% (cross-check)
Risk-free rate (1-year CAL)	4.30%	YTM of Alpha (case); 8-year spot 3.90% used for the 2030 projection
Risk aversion A	4 (tested 2–8)	Team choice; at A = 4 the utility optimum is just inside the stress limit
Trustees' "~15% loss"	stress ≤ 15%, VaR ≤ 10%	Net of payments; stress = equity −40% (≈ 2008) + 5y yield +247 bp (95th pct, 2015–22)
Weight limits (growth)	stock 1–8%, ETF ≤ 35%, sector ≤ 15%	Diversification, every candidate held, liquidity
Costs / cash	10 bp stocks, 3 bp ETFs, bonds at quotes; USD 100k cash at 0%	Team estimate; conservative cash carry

6. Evaluation and rebalancing rule (frozen; run only in the final submission)

This section outlines the evaluation/rebalancing rules applied to the strategy (frozen; run only once within the final submission):

- Bonds will be held until maturity and will then be released. Any future coupon payments for bonds held to maturity (December 2025 & December 2026) will be allocated into the growth portfolio at frozen weights. There will be no change of assets.
- Only when there has been a weight shift greater than three percent points between the last trading date of December 2023 and December 2024 will the growth portfolio be rebalanced back to its original weights. Costs incurred due to rebalancing will be calculated and reflected in turnover.
- A comparison metric will be created utilizing a buy-and-hold (total return) comparator SPY that utilized the exact same beginning value as the strategy.

Metrics used in evaluating the performance of the strategy between 2023-2025 include: cumulative return, annualized return, volatility, Sharpe Ratio, maximum draw down, tracking error, Information Ratio, CAPM Alpha/Beta versus SPY, Fama-French Three Factor Model Alpha/Beta and Carhart four factor model alpha. Real fund value will be tracked versus the USD $8 million dollar path.

Review triggers will include: 1.) An inflation forecast exceeding 3.4 percent or; 2.) A European Central Bank Expected Real Policy Interest Rate forecast below 2.5 percent (probability goal <50%) or; 3.) A stress loss of fifteen percent or higher.

7. Files and reproducibility

The seventh section lists files that will be provided for reproducibility purposes and explains how they can be accessed.
The seven items listed below represent notebooks and directories that contain files related to this project.
These are:

1. Notebooks/01_data_retrieval.ipynb (Downloaded from dated source records + Hashes)
2. Notebooks/02_kappa_analysis.ipynb (Run All; Offline; From Data/Raw)
3. Outputs/original_22Oct (Tables T01-T15; Figures; Frozen_Policy.Json)
4. Docs/Decision_Log.md
5. Data downloaded October 22nd, 2026 (All UTC times recorded in Data/Raw/source_record.csv).